

ID:

Name:

Quiz-1; Time: 10 minutes
Course Title: Statistics and Probability
Course Code: STA102 (10)

Q1. Which of the following is a qualitative variable?

- Ⓐ Age
- Ⓑ Temperature
- Ⓒ Blood type
- Ⓓ Height

Q2. A variable measured using a nominal scale can:

- Ⓐ Be meaningfully added and subtracted
- Ⓑ Be ranked from highest to lowest
- Ⓒ Only be classified into categories
- Ⓓ Be measured in equal units

Q3. The entire collection of individuals or objects under study is called the:

- Ⓐ Parameter
- Ⓑ Sample
- Ⓒ Statistic
- Ⓓ Population

Q4. Which of the following would best represent continuous data?

- Ⓐ Number of books
- Ⓑ Number of children
- Ⓒ Weight in kilograms
- Ⓓ Type of car

Q5. Which graph is most appropriate for categorical data?

- Ⓐ Histogram
- Ⓑ Pie chart
- Ⓒ Scatter plot
- Ⓓ Line graph

Q6. A distributed system records response times (ms): 12, 15, 14, 13, 16, 18, 17, 200. Which measure best represents the *typical* server response time?

- Ⓐ Mean
- Ⓑ Median
- Ⓒ Mode
- Ⓓ Midrange

Q7. If 9th decile response time is 220 ms, it means:

- Ⓐ 9% of responses are exactly 220 ms
- Ⓑ 10% of responses exceed 220 ms
- Ⓒ 90% of responses exceed 220 ms
- Ⓓ Median is 220 ms

Q8. For memory usage logs data: Mean = 48 GB, Median = 38 GB and Mode = 35 GB. The skewness is:

- Ⓐ Positive
- Ⓑ Negative
- Ⓒ Zero
- Ⓓ Cannot be determined

Q9. Two sorting algorithms are tested (in ms):

Algorithm A: 19, 20, 21, 22, 23

Algorithm B: 15, 16, 21, 21, 150

Which statement is correct?

- Ⓐ Both algorithms have equal performance
- Ⓑ Algorithm B is more stable
- Ⓒ Mean of B is heavily influenced by an outlier
- Ⓓ Median of A is larger than B

Q10. Quartiles divide the data into:

- Ⓐ 2 equal parts
- Ⓑ 3 equal parts
- Ⓒ 4 equal parts
- Ⓓ 5 equal parts